

Building Technology

Engineering Mathematics I for Building Technology

Professional Learner Notes

Algebra, trigonometry, mensuration, graphs and introductory calculus for construction practice.

Section	Purpose
Algebra in building works	Formula substitution and transposition.
Trigonometry	Roof pitch, staircase slope and right-triangle applications.
Mensuration of solids	Prisms, cylinders, cones and pyramids.
Graphs and variation	Linear and non-linear relationships.
Introductory calculus	Rate of change and area concepts.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Straight line

$$y=mx+c$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Tangent ratio

$$\tan\theta=\frac{\mathrm{rise}}{\mathrm{run}}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Pyramid volume

$$V=\frac{1}{3}Ah$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Circumference

$$C=\pi d$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Derivative

$$\frac{dy}{dx}=f'(x)$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Algebra in building works

Formula substitution and transposition.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$y=mx+c$$

Worked method

Example: Roof pitch. Find the roof angle when rise is 1.8 m and horizontal span is 4.2 m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports roof pitch. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving excavation volumes. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving cost curves. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving construction graphs. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving roof pitch. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving stair gradients. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving excavation volumes. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving cost curves. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving construction graphs. Show formula, substitution, units and interpretation.

2. Trigonometry

Roof pitch, staircase slope and right-triangle applications.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\tan\theta = \frac{\text{rise}}{\text{run}}$$

Worked method

Example: Excavation. Calculate the volume of a trench using length, width and depth.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports stair gradients. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
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- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving cost curves. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving construction graphs. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving roof pitch. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving stair gradients. Show formula, substitution, units and interpretation.
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6. Create and solve one technical calculation involving cost curves. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving construction graphs. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving roof pitch. Show formula, substitution, units and interpretation.

3. Mensuration of solids

Prisms, cylinders, cones and pyramids.

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- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$V = \frac{1}{3}Ah$$

Worked method

Example: Cost graph. Interpret the gradient and intercept of a labour-cost graph.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports excavation volumes. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

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Practice set

1. Create and solve one technical calculation involving construction graphs. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving roof pitch. Show formula, substitution, units and interpretation.
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4. Graphs and variation

Linear and non-linear relationships.

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Worked method

Example: Roof pitch. Find the roof angle when rise is 1.8 m and horizontal span is 4.2 m.

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Engineering application

This method supports cost curves. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

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Practice set

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5. Introductory calculus

Rate of change and area concepts.

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Worked method

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Mixed Revision Questions

1. A building technology problem involves stair gradients. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A building technology problem involves excavation volumes. Choose a suitable mathematical method, solve the problem, and explain the result.
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References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.